Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Among the following the state function(s) is (are)
(i) Internal energy
(ii) Irreversible expansion work
(iii) Reversible expansion work
(iv) Molar enthalpy

(A) (ii) and (iii)

(B) (i), (ii) and (iii)

(C) (i) and (iv)

(D) (i) only

Q.2

A well stoppered thermos flask contains some ice cubes. This is an example of

(A) Closed system

(B) Open system

(C) Isolated system

(D) Non-thermodynamic system

Q.3

Identify the intensive quantity from the following

(A) Enthalpy and temperature

(B) Volume and temperature

(C) Enthalpy and volume

(D) Temperature and refractive index

Q.4

Which of the following is not a state function

(A) Internal energy

(B) Enthalpy

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(C) Work

(D) Entropy

Q.5

One mole of an ideal gas at $300\ \mathrm{K}$ is expanded isothermally from an initial volume of 1 litre to 10 litres. The $\Delta \mathrm{E}$ for this process is $\left(\mathrm{R}=2 \mathrm{cal}\ \mathrm{mol}^{-1}\ \mathrm{K}^{-1}\right)$

(A) $163.7 \mathrm{cal}$

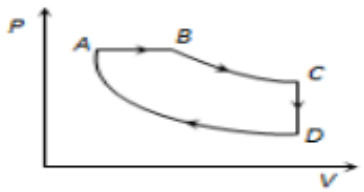
(B) zero

(C) $1381.1 \mathrm{cal}$

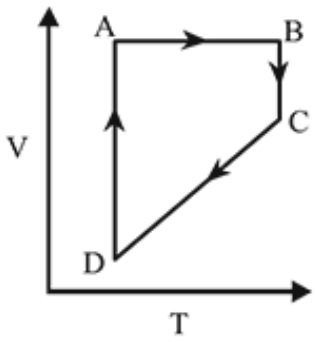
(D) 9 lit atm

Q.6

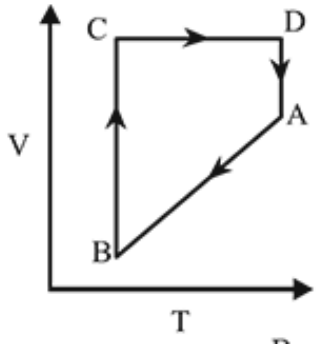
A cyclic process $ABCD$ is shown in $P-V$ diagram for an ideal gas. Which of the following diagram represents the same process?



(A)

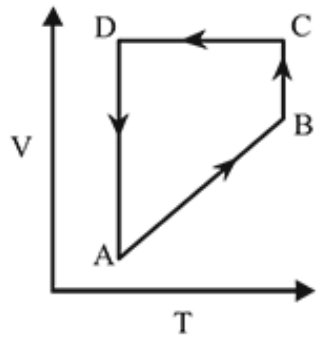


(B)

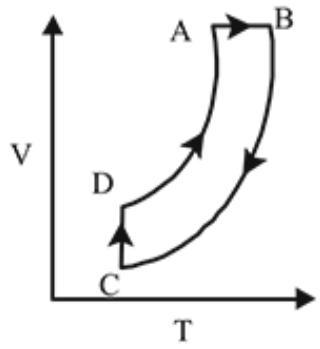


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(C)



(D)



Q.7

Which of the following statements/relationships is not correct in thermodynamic changes?

(A) $\Delta U = 0$ (isothermal reversible expansion of a gas)

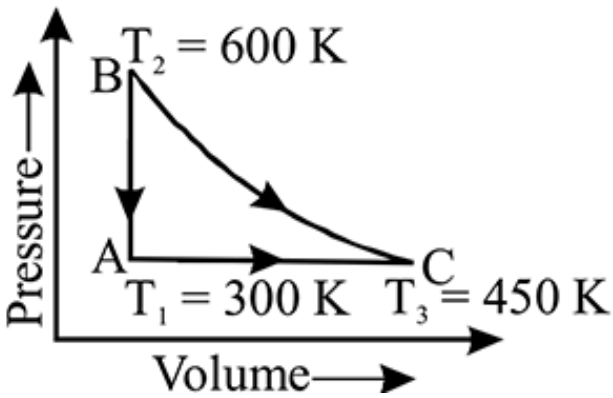
(B) $w = -nRT \ln \frac{V_2}{V_1}$ (isothermal reversible expansion of an ideal gas)

(C) $w = nRT \ln \frac{V_2}{V_1}$ (isothermal reversible expansion of an ideal gas)

(D) For a system of constant volume heat involved directly changes to internal energy.

Q.8

A heat engine carries one mole of an ideal mono-atomic gas around the cycle as shown in the figure. Select the correctoption:



(A) $q_{AB} = 450R$ and $q_{CA} = -450R$

(B) $q_{AB} = 450R$ and $q_{CA} = -225R$

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(C) $q_{AB} = 450R$ and $q_{CA} = -375R$

(D) $\mathrm{q_{AB}} = 375\mathrm{R}$ and $\mathrm{q_{CA}} = -450\mathrm{R}$

Q.9

A gas undergoes change from state A to state B. In this process, the heat absorbed and work done by the gas is $5\ \mathrm{J}$ and $8\ \mathrm{J}$, respectively. Now gas is brought back to A by another process during which $3\ \mathrm{J}$ of heat is evolved. In this reverse process of B to A :

(A) $10\ \mathrm{J}$ of the work will be done by the gas.

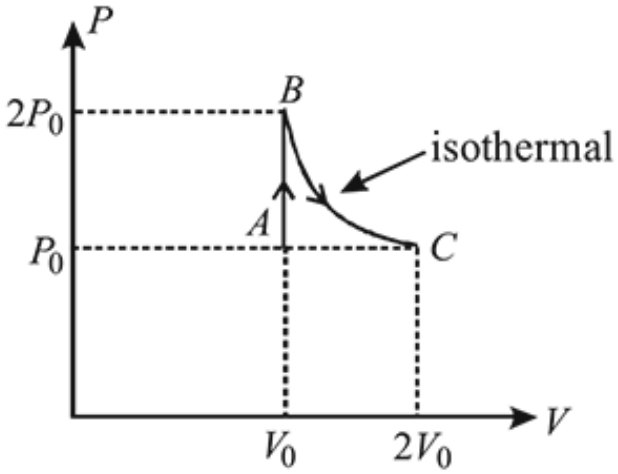
(B) $6\ \mathrm{J}$ of the work will be done by the gas.

(C) $10\ \mathrm{J}$ of the work will be done by the surrounding on gas.

(D) $6\ \mathrm{J}$ of the work will be done by the surrounding on gas.

Q.10

A diatomic ideal gas undergoes a thermodynamic change according to the P-V diagram shown in the figure. The total heat given to the gas is nearly
(use $\ln 2 = 0.7$)



(A) $2.5P_0V_0$

(B) $1.4P_0V_0$

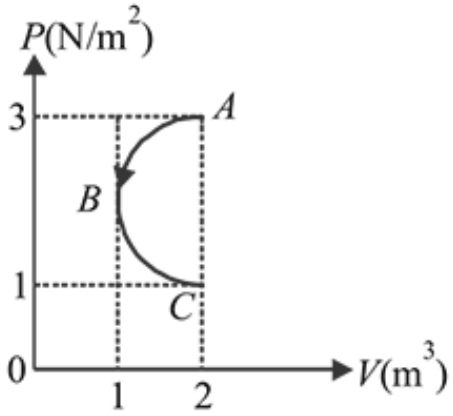
(C) $1.1P_0V_0$

(D) $3.9P_0V_0$

Q.11

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In $P-V$ diagram shown in figure ABC is a semicircle. The work done in the process ABC is



(A) $4\ \mathrm{J}$

(B) $\frac{-\pi}{2}\ \mathrm{J}$

(C) $\frac{\pi}{2}\ \mathrm{J}$

(D) Zero

Q.12

An ideal gas can be expanded from an initial state to a certain volume through two different processes, (A) $\mathrm{PV}^2 = \mathrm{K}$ and (B) $\mathrm{P} = \mathrm{KV}^2$, where K is a positive constant. Then, choose the correct option from the following.

(A) Final temperature in (A) will be greater than in (B)

(B) Final temperature in (B) will be greater than in (A)

(C) Work done by the gas in both the processes would be equal

(D) Total heat given to the gas in (A) is greater than in (B)

Q.13

$1\mathrm{gm}$ of water at a pressure of $1.01 \times 10^5\ \mathrm{Pa}$ is converted into steam without any change of temperature. The volume of $1\ \mathrm{g}$ of steam is 1671 cc and the latent heat of evaporation is $540\mathrm{cal}$. The change in internal energy due to evaporation of $1\mathrm{gm}$ of water is

(A) $\approx 167\mathrm{cal}$

(B) $\approx 500\mathrm{cal}$

(C) $540\mathrm{cal}$

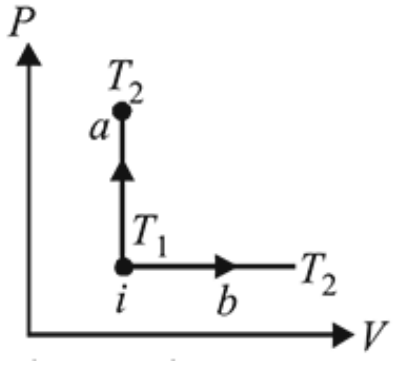
(D) $581\mathrm{cal}$

Q.14

An ideal gas has temperature T_1 at the initial state i shown in the $P-V$ diagram. The gas has a higher

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temperature T_2 at the final states a and b, which it can reach the paths shown. The change in entropy:



(A) greatest in a

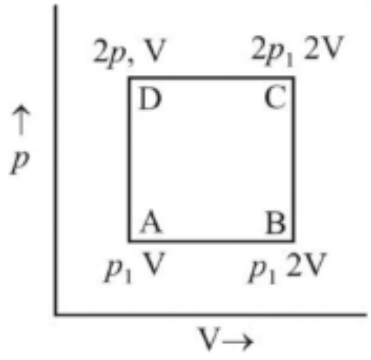
(B) greatest in b

(C) same in a and b

(D) nothing can be said

Q.15

An ideal monatomic gas is taken round the cycle ABCD as shown in the fig. The work done during the cycle is



(A) pV

(B) $2pV$

(C) $\frac{1}{2}pV$

(D) zero

Q.16

Temperature of one mole of helium gas is increased by $1^{\circ}\mathrm{C}$, hence increase in intemal energy is

(A) 7cal

(B) 5cal

(C) 3.5ca1

(D) 3cal

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Q.17

How much energy must be supplied to change $36\ \mathrm{g}$ of ice at $0^{\circ}\mathrm{C}$ to water at room temperature, $25^{\circ}\mathrm{C} \quad \mathrm{AH}^{\circ}$ fusion $= 6.01\ \mathrm{kJ\ mol}^{-1} \quad \mathrm{C}_{\mathrm{p_2\ liquid}} = 4.18\mathrm{JK}^{-1}\ \mathrm{g}^{-1}$

(A) $\mathbf{12kJ}$

(B) $16\ \mathrm{kJ}$

(C) $\mathbf{19kJ}$

(D) $22\ \mathrm{kJ}$

Q.18

Molar heat capacity of water in equilibrium with ice at constant pressure is

(A) Zero

(B) Infinity (∞)

(C) $40.45\ \mathrm{kJ\ K}^{-1}\ \mathrm{mol}^{-1}$

(D) $75.48\ \mathrm{J\ K}^{-1}$

Q.19

The relation between U, P and V for an ideal gas in an adiabatic process is given by relation $U = a + bPV$. Find the value of adiabatic exponent (γ) of this gas

(A) $\frac{b+1}{b}$

(B) $\frac{b+1}{a}$

(C) $\frac{a+1}{b}$

(D) $\frac{a}{a+b}$

Q.20

A Carnot engine whose efficiency is 40%, receives heat at $500\ \mathrm{K}$. If the efficiency is to be 50%, the source temperature for the same exhaust temperature is

(A) $900\ \mathrm{K}$

(B) $600\ \mathrm{K}$

(C) $700\ \mathrm{K}$

(D) $800\ \mathrm{K}$

Answers & Solutions

Q.1 Answer:

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(i) and (iv)

Solution:

Internal energy and molar enthalpy are state functions. Work (reversible or irreversible) is a path function.

Q.2 Answer:

Isolated system

Solution:

An isolated system is a thermodynamic system that can not exchange either energy or matter outside the boundaries of the system. Thus, a well stopped thermos flask containing ice an example for isolated system.

Q.3 Answer:

Temperature and refractive index

Solution:

No solution available

Q.4 Answer:

Work

Solution:

Work is not a state function as during a process its value depends on the path followed. The value of enthalpy, internal energy and entropy depends on the state and not on the path followed to get that state, hence these are state functions.

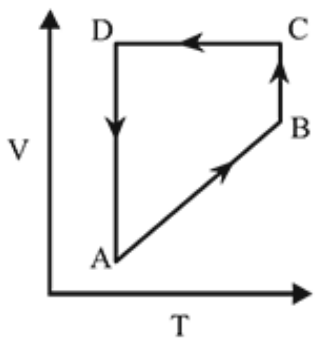
Q.5 Answer:

zero

Solution:

For a finite change $\Delta E = C_v \Delta T$. **For an isothermal process** $\Delta T = 0$. **Hence** $\Delta E = 0$

Q.6 Answer:



Solution:

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AB : Isobaric expansion BC : Isothermal expansion

CD : Isochoric DA : Isothermal compression

Q.7 Answer:

$\mathrm{w}=\mathrm{nRT} \ln \frac{\mathrm{V}_2}{\mathrm{V}_1}$ (isothermal reversible expansion of an ideal gas)

Solution:

For isothermal reversible expansion. $w=-nRT \ln \frac{V_2}{V_1}$

Q.8 Answer:

$q_{AB}=450R$ and $q_{CA}=-375R$

Solution:

For monoatomic gas $\mathrm{C_v}=\frac{3}{2}\mathrm{R} \mathrm{C_p}=\frac{5}{2}\mathrm{R}$

At constant volume, $\Delta \mathrm{U}=\mathrm{q_v}=\mathrm{nC_{v,m}} \Delta \mathrm{T}$

At constant pressure, $\Delta \mathrm{H}=\mathrm{q_p}=\mathrm{nC_{p,m}} \Delta \mathrm{T}$

Q.9 Answer:

$6\ \mathrm{J}$ of the work will be done by the surrounding on gas.

Solution:

$A \overset{q=+5, w=-8J}{\uparrow} B$

$\Delta \mathrm{U_{AB}}=\mathrm{q}+\mathrm{w}=+5+(-8)=-3$

$q=-3, \Delta \mathrm{U_{BA}}=+3$

$\Delta \mathrm{U_{BA}}=\mathrm{q}+\mathrm{w}$

$\Rightarrow 3=-3+\mathrm{w} \Rightarrow \mathrm{w}=+6\ \mathrm{J}$ **(work done on the system).**

Q.10 Answer:

$3.9P_0V_0$

Solution:

$Q_{AB}=\Delta U_{AB}+W_{AB}$

$W_{AB}=0$

$\Delta U_{AB}=\frac{f}{2}nR\Delta T \quad \Rightarrow \quad \frac{f}{2}(\Delta PV)$

$\Delta U_{AB}=\frac{5}{2}(\Delta PV) \Rightarrow Q_{AB}=2.5P_0V_0$

Process BC **:**

$Q_{BC}=\Delta U_{BC}+W_{BC}=0+2P_0V_0 \ln 2=1.4P_0V_0$

$Q_{\text{net}}=Q_{AB}+Q_{BC}=3.9P_0V_0$

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Q.11 Answer:

$\frac{\pi}{2}$ J

Solution:

Net work done is positive and area between semicircle which is equal to $\frac{\pi}{2}(2-1) \times \frac{(3-1)}{2} \mathrm{Nm} = \frac{\pi}{2}$ J

As the curve is going anti-clockwise, the work done is positive.

Q.12 Answer:

Final temperature in (B) will be greater than in (A)

Solution:

Process

(A) $\frac{nRT}{V} V^2 = K; \mathrm{TV} = \text{constant};$

$T \propto \frac{1}{V}$

Process (B) $\frac{nRT}{V} = KV^2; \frac{T}{V^3} =$ **constant;**

$\mathrm{T} \propto \mathrm{V}^3$

Q.13 Answer:

$\approx 500\mathrm{cal}$

Solution:

$$\mathrm{dW} = \mathrm{P}\Delta\mathrm{V} = 1.01 \times 10^5 [1671 - 1] \times 10^{-6} \mathrm{Joule}$$

$$= \frac{1.01 \times 167}{4.2} \mathrm{cal} = 40\mathrm{cal} \text{ nearly}$$

$$\Delta\mathrm{Q} = \mathrm{mL} = 1 \times 540,$$

$$\Delta\mathrm{Q} = \Delta\mathrm{W} + \Delta\mathrm{U}$$

$$\text{or } \Delta\mathrm{U} = 540 - 40 = 500\mathrm{cal}.$$

Q.14 Answer:

greatest in b

Solution:

$Q_a = \Delta U + 0 = \Delta U$ **and** $Q_b = \Delta U + P\Delta V$

As $Q_b > Q_a$

$\therefore$ Change in entropy is greater in case (b).

Q.15 Answer:

pV

Solution:

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The work done during the cyclic path is equal to the area enclosed by the path, which is,
pV.

Q.16 Answer:

3cal

Solution:

$\left(\frac{\Delta E}{\Delta T}\right)_V = C_v$
$\therefore \mathbf{\Delta E} = \mathrm{Cv}\mathbf{\Delta T}$
$\mathrm{C_v} = \frac{3}{2}\mathrm{R}$ **for monoatornic gas**
$\Delta \mathrm{T} = 1^\circ$
$\therefore \Delta \mathrm{E} = \frac{3}{2}\mathbf{R} = \frac{3}{2} \times 2 = 3\mathrm{cal}$

Q.17 Answer:

16 kJ

Solution:

$\Delta H = \Delta H$ (36 g **ice at** $0^\circ\mathrm{C}$ **to 36** $\mathrm{gH_2O}$ **at** $0^\circ\mathrm{C}$) + $\triangle \mathrm{H}$ (36 $\mathrm{gH_2O}$ **at** $0^\circ\mathrm{C}$ **to 36** $\mathrm{gH_2O}$ **at** $25^\circ\mathrm{C}$)
$= \mathbf{\Delta H_{furion}} \times \mathbf{mol} + \mathbf{C_p x}$ **mass** $\mathbf{x \Delta T}$
$= \left(6.01 \times \frac{36}{18}\right) + \frac{4.18}{1000} \times 36 \times 25$
kJ k
$= 15.782 \approx 16\ \mathrm{kJ}$

Q.18 Answer:

Infinity (∞)

Solution:

$\mathrm{C_P} = \left(\frac{\delta \mathrm{H}}{\delta \mathrm{T}}\right)_\mathrm{P}$, **At equilibrium** T **is constant i.e.** $\delta \mathrm{T} = 0$,
thus $C_\mathrm{P} = \infty$

Q.19 Answer:

$\frac{b+1}{b}$

Solution:

$U = a + bPV$
In adiabatic change,
$dU = -dW = \frac{nR}{\gamma - 1}(T_2 - T_1) = \frac{nR}{\gamma - 1}(dT)$
$\Rightarrow U = \int dU = \frac{nR}{\gamma - 1}\int dT$
or $\quad U = \left(\frac{nR}{\gamma - 1}\right)T + a = \frac{PV}{\gamma - 1} + a \ldots\ldots (2)$

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where a is the constant of integration.

Comparing (1) and (2), we get $b = \frac{1}{\gamma - 1} \Rightarrow \gamma = \frac{b+1}{b}$.

Q.20 Answer:

$600\ \mathrm{K}$

Solution:

Efficiency of Carnot engine, $\eta = 1 - \frac{\mathrm{T}_2}{\mathrm{T}_1}$ where T_1 and T_2 be the temperature of source and sink respectively. $\therefore \quad \frac{\mathrm{T}_2}{\mathrm{T}_1} = 1 - \eta = 1 - \frac{40}{100} = \frac{60}{100} = \frac{3}{5} \quad (\because \eta = 40\%)$

$\mathrm{T}_2 = \frac{3}{5}\,\mathrm{T}_1 = \frac{3}{5} \times 500\ \mathrm{K} = 300\ \mathrm{K}$

$(\because T_1 = 500\ \mathrm{K})$

Let T_1' be the temperature of the source for the same sink temperature when efficiency $\eta' = 50\%$

$\therefore \frac{\mathrm{T}_2}{\mathrm{T}_1'} = 1 - \eta' = 1 - \frac{50}{100} = \frac{1}{2}$

$\mathrm{T}_1' = 2\,\mathrm{T}_2 = 2 \times 300\ \mathrm{K} = 600\ \mathrm{K}$ **(Using eq.**

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